JACC VOL. ■, NO. ■, 2026

PUBLISHED BY ELSEVIER

Sex and Age Specific Genetic Risk Across the Dilated and Arrhythmogenic Cardiomyopathy Spectrum

Insights From the SHaRe Registry

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ABSTRACT

BACKGROUND Dilated cardiomyopathy (DCM) and arrhythmogenic cardiomyopathy (ACM) are progressive cardiac muscle disorders with phenotypic and genetic overlap. Although a male predominance is noted in DCM/ACM, it remains unclear whether this extends to specific genetic subtypes or reflects variation in disease stage and whether sex influences age-dependent disease onset across pediatric and adult groups.

OBJECTIVES The aim of this study was to define sex-based differences in genetic architecture and age at diagnosis of DCM/ACM across pediatric and adult populations.

METHODS Genetically tested adult and pediatric DCM/ACM patients and asymptomatic genotype-positive relatives enrolled in the multicenter SHaRe (Sarcomeric Human Cardiomyopathy Registry) were analyzed. Sex distribution across 27 DCM- and ACM-associated genes was evaluated using logistic regression. Age at diagnosis was compared across sex and genes using Kaplan-Meier cumulative incidence estimates.

RESULTS Among 3,410 patients, a 61% male predominance was present across subgroups of genotype positive, genotype negative, and variants of uncertain significance ($P = 0.008$), with significant gene-specific variation. *TTN* truncating variants (TTNtv) were less common in females (OR: 0.42 [95% CI: 0.33-0.54]; $P < 0.01$), while *DSP* (OR: 3.3 [95% CI: 2.35-4.78]; $P < 0.01$) and grouped non-*TTN* sarcomeric variants (*ACTC1*, *TNNT2*, *MYH7*, *TNNC1*, *TNNI3*, and *TPM1*) were more common (OR: 1.68 [95% CI: 1.15-2.47]; $P < 0.001$) in females compared with males with DCM/ACM. Age at diagnosis was comparable between sexes, except in TTNtv carriers, among whom males exhibited earlier disease onset compared with females (median age 45 years [Q1-Q3: 33-55 years] vs 51 years [Q1-Q3: 38-60 years]; $P = 0.003$). Pediatric-onset (diagnosis at <18 years) cases (n = 174) also demonstrated a male predominance (60% male) but had distinct genetic characteristics, with non-*TTN* sarcomeric and *PKP2* variants being more prevalent compared with adult-onset disease (non-*TTN* sarcomeric OR: 5.5 [95% CI: 3.3-8.9; $P < 0.01$]; *PKP2* OR: 2.8 [95% CI: 1.1-5.2; $P < 0.05$]) and a bimodal age at onset peaking in infancy and adolescence.

CONCLUSIONS Gene-specific sex differences influence disease prevalence and age at onset in DCM/ACM. TTNtv are more common with earlier onset in males, whereas *DSP* and non-*TTN* sarcomeric variants predominate in females. Pediatric-onset DCM/ACM is genetically distinct and caused predominantly by non-*TTN* sarcomeric variants, especially during infancy. These findings support age- and sex-informed surveillance strategies and prioritize future research into the mechanisms of observed sex-based differences. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

ISSN 0735-1097/$36.00 https://doi.org/10.1016/j.jacc.2026.03.054

ABBREVIATIONS AND ACRONYMS

ACM = arrhythmogenic cardiomyopathy

ARVC = arrhythmogenic right ventricular cardiomyopathy

BMI = body mass index

CMR = cardiac magnetic resonance imaging

DCM = dilated cardiomyopathy

HCM = hypertrophic cardiomyopathy

G+/P− = genotype positive/phenotype negative

NDLVC = nondilated left ventricular cardiomyopathy

P/LP = pathogenic or likely pathogenic

TTNtv = *TTN* truncating variant

VUS = variant(s) of uncertain significance

Dilated cardiomyopathy (DCM) and arrhythmogenic cardiomyopathy (ACM) are progressive primary myocardial disorders that share an overlapping phenotypic spectrum and frequently have a genetic basis, with pathogenic or likely pathogenic (P/LP) variants identified in approximately 20% to 40% of cases.[1,2]

Epidemiologic data suggest a male predominance in DCM/ACM, which is not due solely to underdiagnosis in females.[3-5] In DCM, males are more frequently affected and exhibit worse left ventricular function and dimensions.[6] Similarly, in arrhythmogenic right ventricular cardiomyopathy (ARVC), male sex is associated with higher penetrance, increased arrhythmic burden, and more severe structural progression.[5] However, it remains unclear whether this male predominance is driven by differences in disease stage at presentation and whether it extends to a gene-specific level.

Although cardiomyopathy causing *TTN* truncating variants (TTNtv) have been reported as more prevalent in males, limited studies have systematically evaluated sex differences in other genetic subtypes, such as *FLNC*, and sarcomeric genes such as *MYH7* and *TNNT2*.[7,8] We have recently demonstrated sex differences in the genetic architecture of DCM,[9] identifying distinct patterns of variant burden between males and females in a population-based framework. However, that work did not address whether these genetic differences reflect variation in disease stage or clinical presentation at diagnosis. In the present study, using a substantially larger multicenter registry of genotyped DCM/ACM patients across the age spectrum, we confirm and extend these observations by incorporating detailed phenotyping and age-at-onset analyses, including ACM and nondilated left ventricular cardiomyopathy (NDLVC) phenotypes and pediatric-onset architecture, to determine whether sex-associated genotype patterns persist independent of clinical context and disease stage. By integrating genomic and phenotypic data, we aimed to dissect the biological vs clinical contributors to sex differences in DCM, with the goal of identifying gene-specific patterns that may inform precision medicine strategies. Leveraging the multicenter, international SHaRe (Sarcomeric Human Cardiomyopathy Registry), we aimed to systematically assess sex- and gene-specific differences in the age at onset of genetic DCM/ACM across a large cohort of genotyped individuals. Given the overlapping disease spectrum of DCM/ACM, we analyzed these cardiomyopathies together while accounting for gene-specific differences. Specifically, we investigated: 1) sex differences across genetic subtypes, identifying whether the male predominance in DCM/ACM is present in disease caused by individual genes; and 2) age at diagnosis as a surrogate for penetrance, assessing whether males and females develop clinical disease at different ages within distinct genetic subgroups. In addition, we evaluated differences in genetic distribution between adult-onset and pediatric-onset disease. By addressing these gaps, this study provides new insights into sex-based genetic susceptibility in DCM/ACM, with implications for future research, genetic counseling, risk stratification, and screening guidelines.

METHODS

PARTICIPATING SITES AND CREATION OF THE CENTRALIZED DATABASE. SHaRe was established by experienced, high-volume DCM and ACM centers that maintain longitudinal databases capturing

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Manuscript received November 21, 2025; revised manuscript received March 9, 2026, accepted March 12, 2026.

clinical baseline and genetic data on patients with DCM/ACM, across the age spectrum, including pediatric patients (<18 years of age) and family members. In this study, SHaRe refers to the SHaRe DCM/ACM registry and is distinct from the SHaRe hypertrophic cardiomyopathy (HCM) registry.[10] Definitions were harmonized for key demographic, clinical, and genetic parameters. Site data (without protected health information) were mapped to 762 discrete corresponding fields in the secure centralized database to yield standardized SHaRe definitions and values (Boston Advanced Analytics). This international study analyzed data from 11 different sites: Brigham and Women's Hospital (Boston, Massachusetts); Stanford University (Stanford, California); Cardiomyopathy Unit, University of Florence (Florence, Italy); Maastricht University Medical Center (Maastricht, the Netherlands); Royal Brompton Hospital (London, United Kingdom); Colorado University Medical Center (Aurora, Colorado); the University of Michigan (Ann Arbor, Michigan, USA); Northwestern Medicine (Chicago, Illinois); Hospital of the University of Pennsylvania (Philadelphia, Pennsylvania); the Children's Hospital of Philadelphia (Philadelphia, Pennsylvania); and the University Hospital of Trieste (Trieste, Italy) (Supplemental Figure 1). Prospective data were captured via quarterly uploads from site databases. Institutional Review Board and ethics approval was obtained in accordance with policies applicable to each SHaRe site. All participants provided written informed consent.

Primary data cannot be made available to others for the purposes of reproducing or replicating the procedure because of the constraints related to site-specific ethical approvals. Analytical methods can be made available on request.

STUDY POPULATION. Patients with primary DCM or ACM who underwent genetic testing were included in this study. In addition, NDLVC patients were also included under DCM/ACM. ARVC patients were also included in the analysis, although they only constituted a small fraction of the total cohort (n = 206; n = 67 women, n = 139 men). Given the recognized clinical and genetic overlap across left- and right-dominant cardiomyopathy phenotypes, we refer to the combined cohort throughout as DCM/ACM. Data up until January 2025 were included. DCM/ACM patients without an age at diagnosis were excluded (Supplemental Figure 2). Participants excluded for missing age at diagnosis who had undergone genetic testing (n = 210) were predominantly male (69%), and 46% were genotype positive. DCM/ACM diagnosis was determined independently by each participating center by expert clinicians as part of routine clinical care, following international guidelines. Given the recognized phenotypic overlap between DCM and ACM, these conditions were analyzed within a shared disease spectrum. This approach reflects the dynamic nature of cardiomyopathy classification, which continues to evolve with growing insights.

The affected cohort included both probands and affected relatives enrolled in SHaRe. Pediatric-onset DCM/ACM was defined as diagnosis before 18 years of age and was analyzed as a subgroup within the overall cohort. Pediatric-onset cases were stratified into 4 age groups: infancy (<1 year), early childhood (1-6 years), late childhood (6-11 years), and adolescence (11-18 years).

GENETIC TESTING. Genetic testing was initiated at all sites and was performed at participating centers using institution-specific gene panels. Variants were curated by an experienced genetic team at each center and classified as pathogenic, likely pathogenic, P/LP, variants of uncertain significance (VUS), or likely benign or benign according to American College of Medical Genetics and Genomics criteria. We focused on genes that were classified as definite or strong and moderately associated with DCM, ACM, and ARVC per the ClinGen consortium.[11,12] As ClinGen classifications are periodically updated with new evidence, our analyses reflect the ClinGen classifications available at the time the study was designed. The genes included were *BAG3*, *DES*, *DSG2*, *DSC2*, *FLNC*, *JUP*, *LMNA*, *MYH7*, *PKP2*, *PLN*, *RBM20*, *SCN5A*, *TNNC1*, *TNNT2*, *TTN*, *DSP*, *ACTC1*, *ACTN2*, *JPH2*, *NEXN*, *TNNI3*, *TMEM43*, *TPM1*, and *VCL*. Panel composition varied over time and among sites. Genetic testing occurred predominantly after 2010 (7.3% before 2010, 33.4% from 2010 to 2015, 30.2% from 2016 to 2020, and 29.1% after 2020). Genotyped patients were then designated as genotype positive (P/LP+; 1 P/LP variant in any of the aforementioned genes), multiple variants (>1 P/LP variant in the aforementioned genes), VUS+ (VUS present any of the aforementioned genes), and genotype negative (P/LP−; no potentially pathogenic variants identified in the aforementioned genes). *ACTC1*, *TNNT2*, *MYH7*, *TNNC1*, *TNNI3*, and *TPM1* were grouped as sarcomeric variants. The sarcomeric gene *TTN* was grouped individually. *TTN* missense variants were classified locally as class 2 or VUS and were not included in the *TTN* group. Individuals with >1 P/LP in the multiple variants group were excluded from gene-specific analyses. Individuals categorized as VUS carried VUS in the absence of any P/LP variant. VUS were not systematically captured and thus may be underrepresented. Genotype negative does not imply nongenetic but rather that a monogenic cause was not identified using contemporary panel-based testing.

Additionally, family members without phenotypic manifestations of DCM/ACM but with P/LP variants identified in the aforementioned genes were identified (termed genotype positive/phenotype negative [G+/P−]). Family members were included if they had undergone echocardiography or cardiac magnetic resonance imaging (CMR) as part of the hospital's screening program and showed no signs or symptoms of DCM/ACM at first assessment, as described earlier. The age at last follow-up was determined as the date of the most recent echocardiographic or CMR examination in asymptomatic family members. Phenotype conversion of G+/P− individuals to overt DCM/ACM was determined at the site on the basis of interval development of clinical manifestations.

CLINICAL EVALUATION OF PATIENTS WITH DCM/ACM. Initial evaluation of DCM/ACM was defined as the first outpatient clinic visit of a DCM/ACM patient at the medical center that enrolled the patient and/or first imaging using echocardiography or CMR. Where present, we captured the results of physical examination, 12-lead electrocardiography, ambulatory electrocardiography, cardiac imaging (echocardiography and/or CMR), and planned regular follow-ups. Key information collected at the initial visit included the patient's age when diagnosed, NYHA functional classification, any family history of DCM/ACM, and demographic details. Race and ethnicity could not be collected at all participating sites because of national privacy laws. Data on prevalent cardiovascular risk factors such as hypertension, diabetes, and body mass index (BMI) were gathered.

STATISTICAL ANALYSIS. Genetically at-risk family members without phenotypic manifestations of disease (G+/P−) were excluded from the baseline characteristics and age at diagnosis analysis except where otherwise specified. For baseline characteristics, missing data were handled using complete case analysis for the respective variable. Normally distributed data are expressed as mean ± SD and were compared using Student's *t*-test. Non-normally distributed data are expressed as median (Q1-Q3) and were compared using the Mann-Whitney *U* test and the Kruskal-Wallis test, as appropriate. Differences in categorical variables were calculated using the chi-square test or Fisher exact test. To estimate gene-specific ORs, we fit separate logistic regression models within each individual disease gene cohort using sex as the predictor. In each model, carriers of the gene of interest were compared with all noncarriers of that gene (total cohort). Accordingly, an OR >1 indicates that P/LP variants in that gene are more frequent in females than expected on the basis of the overall cohort sex distribution, whereas an OR <1 indicates male predominance. In addition, the OR for pediatric- vs adult-onset disease, dichotomized according to age at onset before 18 years, was determined. The Kaplan-Meier estimates of cumulative incidence with a log-rank test for significance included affected individuals and genotype-positive relatives without a recorded diagnosis at last follow-up, who were treated as censored at their last clinical evaluation. *P* values <0.05 were considered to indicate statistical significance. Statistical analysis was performed using Stata BE version 18.0 (StataCorp) and visualized using Prism version 10.2.0 (GraphPad).

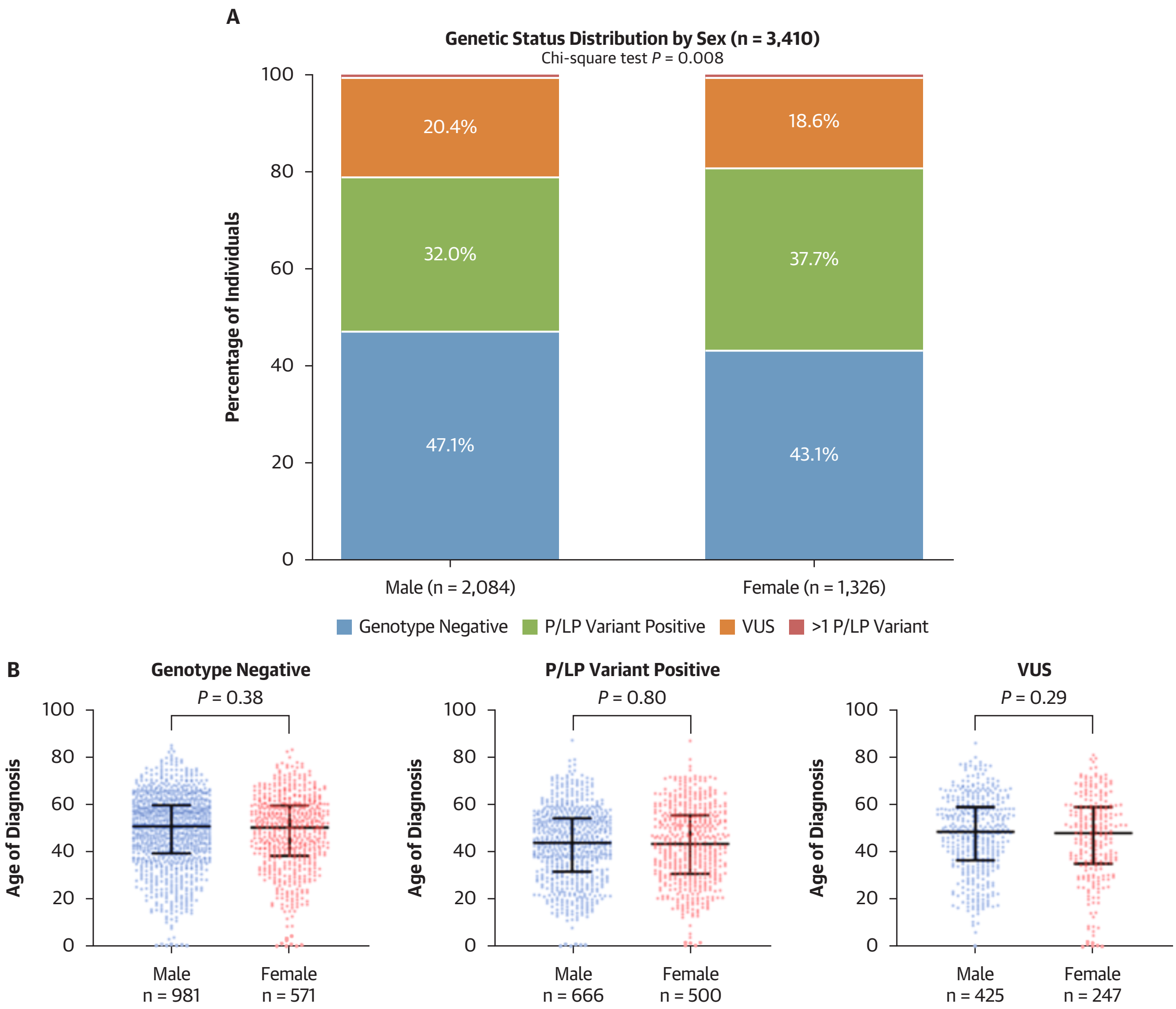
RESULTS

COHORT CHARACTERISTICS AND AGE AT DIAGNOSIS. The study cohort included 3,410 patients with DCM/ACM who underwent genetic testing (P/LP, n = 1,186; VUS, n = 672; genotype negative, n = 1,552), of whom 2,084 (61%) were male. The distribution of genetic status differed by sex (chi-square *P* = 0.008), with females showing a higher proportion of P/LP-positive results and males a higher proportion of genotype-negative results (**Figure 1A**). In the 166 genotype-positive family members without phenotypic disease (G+/P−), we observed a female predominance (n = 119 [72%]; *P* < 0.001).

Several genotypes were observed at very low frequencies in our cohort and are not further discussed in detail: *SCN5A* (n = 23 total; 13 males, 3 pediatric), *RBM20* (n = 20 total; 13 males, 2 pediatric), *BAG3* (n = 16 total; 5 males, 0 pediatric), *PLN* (n = 14 total; 4 males, 1 pediatric), *DSG2* (n = 10 total; 7 males, 1 pediatric), *DES* (n = 11 total; 8 males, 0 pediatric), *NEXN* (n = 6 total; 4 males, 0 pediatric), *DSC2* (n = 4 total; 2 males, 1 pediatric), *TMEM43* (n = 2 total; 2 males, 0 pediatric), *JUP* (n = 2 total; 1 male, 0 pediatric), and *VCL* (n = 1 total; 1 male, 0 pediatric) (Supplemental Table 1, Supplemental Figure 3).

There was no significant difference between the age of DCM/ACM diagnosis in males and females, irrespective of composite genetic status (**Figure 1B**). The median age at diagnosis for males was 48 years (Q1-Q3: 36-58 years), while for females, it was 47 years (Q1-Q3: 34-58 years) (*P* = 0.2) (**Table 1**). Phenotypic data indicated that females with DCM

FIGURE 1 Sex and Age at Onset Across Genetic Subgroups in the DCM/ACM Cohort



(A) Distribution of genetic status per sex in dilated cardiomyopathy (DCM)/arrhythmogenic cardiomyopathy (ACM) patients in percentages. The chi-square test was used to compare the 4 genetic subgroup proportions for males and females. Although included in the legend and analysis, individuals with >1 pathogenic or likely pathogenic (P/LP) variant constituted a very small proportion of the cohort (<1% per sex), and the corresponding segment may appear absent because of scale. (B) Age at DCM/ACM diagnosis per genetic status, stratified by sex. The plots display the median and interquartile range, and each point on the plot represents data from an individual patient. VUS = variant(s) of uncertain significance.

exhibited more ventricular dilatation (echocardiography: left ventricular end-diastolic diameter index 31 ± 8.5 mm/m^2 in females vs 29 ± 6.9 mm/m^2 in males; $P < 0.001$) while presenting with higher left ventricular ejection fractions (42% ± 14.3% in females vs 39% ± 15.0% in males; $P < 0.001$) and right ventricular ejection fractions (52% ± 11.8% in females vs 47% ± 12.8% in males; $P < 0.001$) compared with males (**Table 1**).

GENE-SPECIFIC DIFFERENCES IN SEX DISTRIBUTION. A clear sex difference was observed in the distribution of male and female patients per specific disease gene (**Figure 2**, Supplemental Figure 3, Supplemental Table 2). TTNtv were more prevalent in males compared with female patients. The odds of having a *TTN* P/LP variant were 58% lower in females compared with males (OR: 0.42 [95% CI: 0.33-0.54]; $P < 0.0001$). In contrast, *DSP* variants and sarcomeric

TABLE 1 Baseline Characteristics of the Total Cohort by Sex

	n	Male (n = 2,084)	Female (n = 1,326)	P Value
Age at diagnosis, y	3,410	46.6 ± 16.0	45.8 ± 16.5	0.13
ARVC diagnosis	204	138 (67.6)	66 (32.4)	0.01
Pediatric diagnosis (<18 y)		104 (5.0)	70 (5.3)	0.71
Race	988			
White		928 (88.6)	624 (89.8)	0.12
Black		82 (7.8)	47 (6.8)	
Asian		30 (2.9)	13 (1.9)	
Other		7 (0.7)	11 (1.6)	
Proband[a]	3,054	n = 1,909 (91.6)	1,145 (86.3)	
Genetic status	3,410			0.008
Genotype negative		981 (47.1)	571 (43.1)	
P/LP variant positive		666 (32.0)	500 (37.7)	
Two P/LP variants		12 (0.6)	8 (0.6)	
VUS		425 (20.4)	247 (18.6)	
NYHA functional class	2,800			0.010
I		968 (56.4)	551 (50.8)	
II		549 (32.0)	403 (37.2)	
III/IV		199 (11.6)	130 (12.0)	
Rhythm on baseline ECG	1,813			<0.001
Sinus rhythm		916 (85.3)	663 (89.7)	
Atrial fibrillation or flutter		59 (5.5)	12 (1.6)	
Paced		80 (7.4)	52 (7.0)	
Other		19 (1.8)	12 (1.6)	
Body mass index, kg/m^2	2,647	27.4 ± 5.1	26.8 ± 6.6	0.005
Hypertension	3,410	281 (13.5)	165 (12.4)	0.38
Diabetes	3,410	111 (5.3)	68 (5.1)	0.80
TTE	2,764			
LVEDD, mm		59.0 ± 9.6	53.8 ± 8.5	<0.001
LVEDDi, mm/m^2		29.3 ± 6.9	31.0 ± 8.5	<0.001
LVEF, %		39.8 ± 15.4	42.3 ± 15.2	<0.001
MRI	1,697			
LVEDV, mL		245.2 ± 84.6	189.8 ± 66.4	<0.001
LVEDVi, mL/m^2		119.3 ± 40.2	105.5 ± 36.6	<0.001
RVEDV, mL		189.4 ± 57.2	139.7 ± 43.6	<0.001
RVEDVi, mL/m^2		90.8 ± 26.4	78.2 ± 26.1	<0.001
LVEF, %		39.6 ± 14.7	42.7 ± 13.8	<0.001
RVEF, %		46.8 ± 12.8	51.9 ± 11.8	<0.001
Combined MRI or TTE LVEF, %	3,046	39.1 ± 15.0	42.0 ± 14.3	<0.001

Values are mean ± SD or n (%), unless otherwise indicated. Pediatric-onset cases (diagnosis at <18 years of age) are a subset of the overall affected cohort. [a]Restricted to probands.

ARVC = arrhythmogenic right ventricular cardiomyopathy; ECG = electrocardiography; LVEDD = left ventricular end-diastolic diameter; LVEDDi = left ventricular end-diastolic diameter index; LVEDV = left ventricular end-diastolic volume; LVEDVi = left ventricular end-diastolic volume index; LVEF = left ventricular ejection fraction; MRI = magnetic resonance imaging; P/LP = pathogenic or likely pathogenic; RVEDV = right ventricular end-diastolic volume; RVEDVi = right ventricular end-diastolic volume index; RVEF = right ventricular ejection fraction; TTE = transthoracic echocardiography; VUS = variant of uncertain significance.

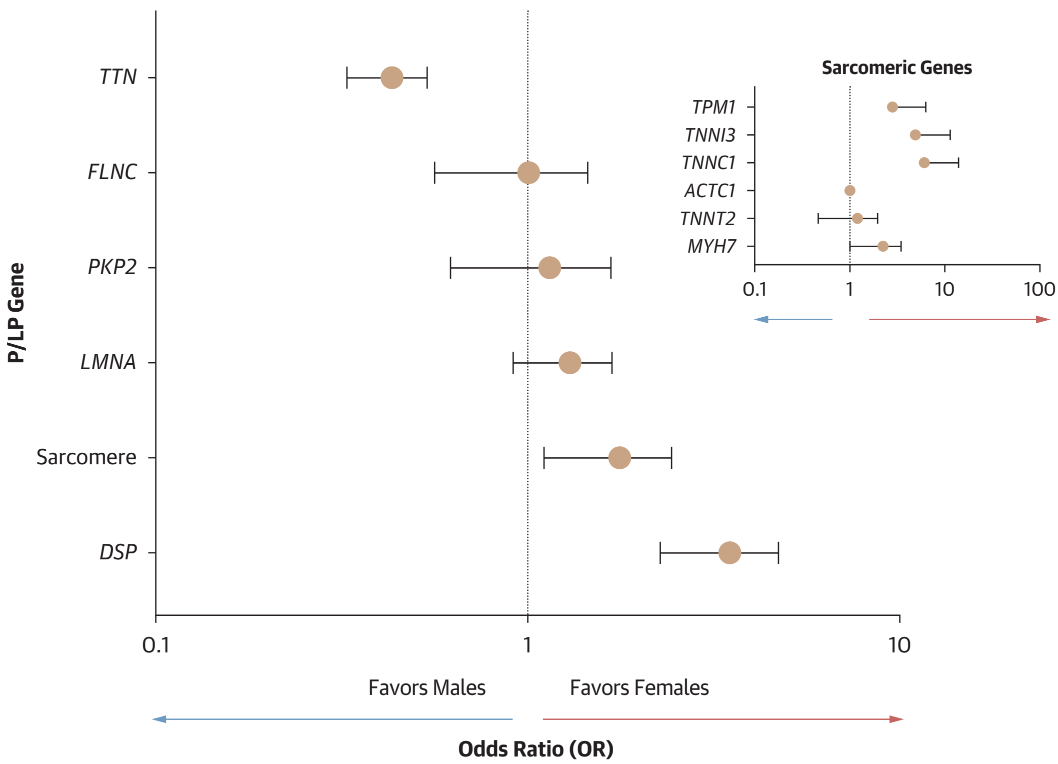
(non-*TTN*) variants showed a female bias, with 3.3-fold (95% CI: 2.35-4.78; $P < 0.0001$) and 1.68-fold (95% CI: 1.15-2.47; $P = 0.007$) higher odds of being present in females compared with males, respectively. *FLNC*, *LMNA*, and *PKP2* did not exhibit a statistically significant sex predominance.

We additionally performed a sensitivity analysis excluding individuals with classically defined ARVC. The exclusion of individuals classified with ARVC (n = 204; 67 females, 139 males) did not materially alter the primary gene-by-sex associations observed in the adult-onset cohort. Male predominance among *TTN* variant carriers persisted (49.6% in males vs 27.0% in females), while female predominance remained evident for DSP (21.7% in females vs 7.0% in males) and non-*TTN* sarcomeric variants (12.0% in females vs 8.6% in males) (Supplemental Table 2). Exclusion of ARVC cases yielded similar gene-by-sex patterns to the primary analysis, supporting the robustness of the primary findings.

AGE AT DIAGNOSIS PER DISEASE GENE. When comparing the age at diagnosis between sexes per disease gene, the age at diagnosis differed in male and female patients with *TTN* P/LP variants (**Figure 3**). Male patients with *TTN* P/LP variants were diagnosed at a younger age, with a median age of 45 years (Q1-Q3: 33-55 years) compared with females, who had a median age of 51 years (Q1-Q3: 38-60 years) ($P = 0.003$). In contrast, no significant age differences at diagnosis were observed between males and females harboring non-*TTN* variants. Age at diagnosis varied across different disease genes in both male and female patients, as shown by significant differences in the distribution of diagnosis ages within each sex ($P = 0.002$ for males; $P < 0.0001$ for females) (Supplemental Figure 4). In males, *FLNC*, *PKP2*, and sarcomeric genes showed a younger age at diagnosis, while in females, *PKP2* and *DSP* showed a younger age at diagnosis. When analyzed by disease gene, we did not identify sex-specific differences in medical history, heart failure severity, or pattern of ventricular remodeling at diagnosis (Supplemental Table 3).

CUMULATIVE DISEASE INCIDENCE. We assessed cumulative disease incidence across affected individuals with the addition of genotype-positive unaffected relatives: G+/P− (censored at last follow-up, n = 166) from 10 SHaRe sites (**Figure 4**, Supplemental Table 4). In the P/LP variant subgroup, males demonstrated earlier diagnosis compared with females (log-rank $P = 0.003$). This was driven by *TTN* variant carriers, among whom males developed DCM/ACM at a younger age than females (log-rank $P = 0.002$). Because relatively few genotype-positive relatives remain under longitudinal surveillance into later decades (often because of censoring at the last screening visit), the curves may visually converge toward 100% despite a minority remaining unaffected. Kaplan-Meier curves truncated at 60 years of

FIGURE 2 **Gene-Specific Sex Bias in DCM/ACM Patients**



Sex-specific ORs and 95% CIs shown for the sex distribution per gene (P/LP variant in underlying gene) in DCM/ACM patients. The dotted vertical line at an OR of 1 represents no sex bias. Genes positioned to the left of the line favor males, while those to the right favor females. The sarcomere category captures all the non-*TTN* sarcomere genes depicted in the top right corner inset. *TTN* truncating variants are more likely to occur in male patients, whereas *DSP* and non-*TTN* sarcomeric variants are more likely to occur in female patients with DCM/ACM. Abbreviations as in **Figure 1**.

age are shown in Supplemental Figure 5 and demonstrate <100% cumulative incidence across all groups. In contrast, cumulative incidence curves for other disease genes did not show significant sex differences, suggesting a more balanced disease age at onset between males and females (Supplemental Figure 6). The cumulative incidence of disease did not significantly vary by disease gene within the only-male and only-female groups ($P = 0.06$ for both sexes) (Supplemental Figure 7).

GENETIC DISTRIBUTION IN PEDIATRIC-ONSET DCM/ACM. A subgroup analysis focusing on pediatric-onset DCM/ACM was conducted, encompassing 174 patients (70% probands) included from 10 SHaRe centers diagnosed before the 18 years of age (**Table 2**). Variants in sarcomeric genes were the most prevalent in pediatric patients (n = 23), followed by *DSP* (n = 12), *LMNA* (n = 12), *TTN* (n = 11), and *PKP2* (n = 10). The distribution of disease-causing genes in pediatric cases differed from the adult population (**Figure 5**, Supplemental Figure 8). *TTN* was less prevalent in pediatric-onset DCM/ACM (OR: 0.26 [95% CI: 0.14-0.50]; $P < 0.0001$). In contrast, non-*TTN* sarcomeric variants and *PKP2* variants were enriched in pediatric cases (non-*TTN* sarcomeric OR: 5.5 [95% CI: 3.3-8.9; $P < 0.01$]; *PKP2* OR: 2.8 [95% CI: 1.1-5.2; $P < 0.05$]). Notably, all pediatric cases with *PKP2* P/LP variants were classified as ARVC.

Genetic status distributions were similar between males and females (chi-square $P = 0.70$) (**Figure 6A**). Differences were observed in the distribution of sex per disease gene. Male predominance was observed for *TTN* (n = 10 [91%]), while female predominance was observed for *DSP* (n = 9 [75%]). In pediatric-onset DCM/ACM, the median age at onset was 15 years (Q1-Q3: 11-17 years) in males and 13 years (Q1-Q3: 2-16 years) in females ($P = 0.03$) (**Figure 6B**).

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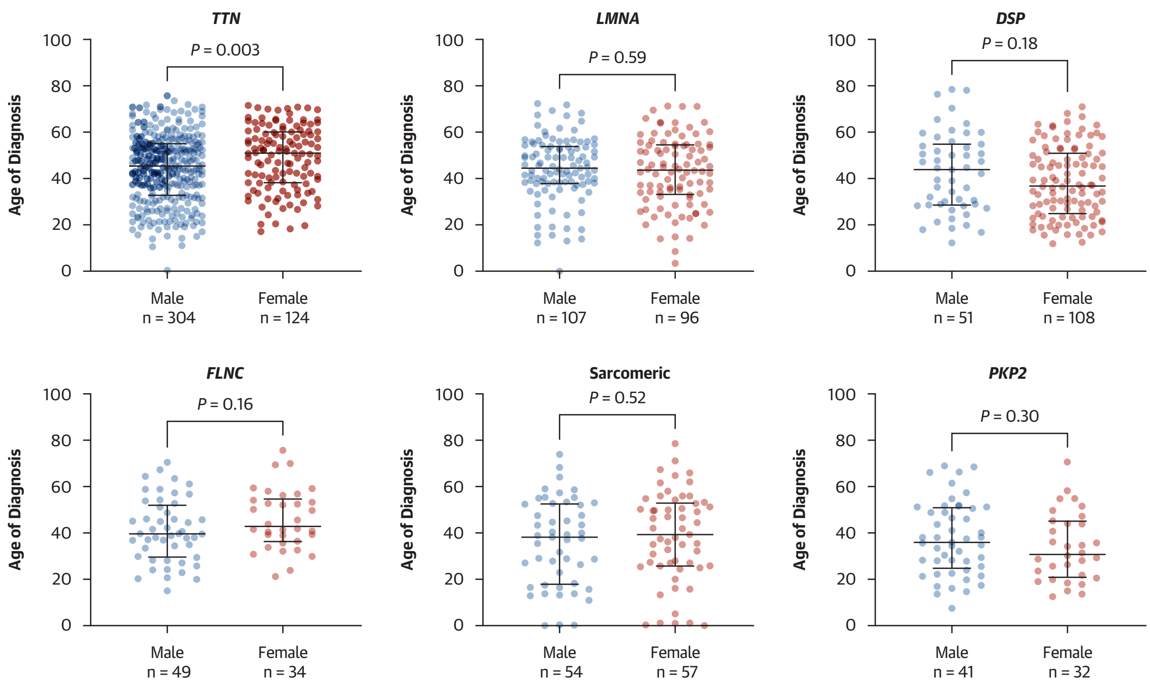
FIGURE 3 Age at DCM/ACM Diagnosis per Gene Group in Male and Female Patients



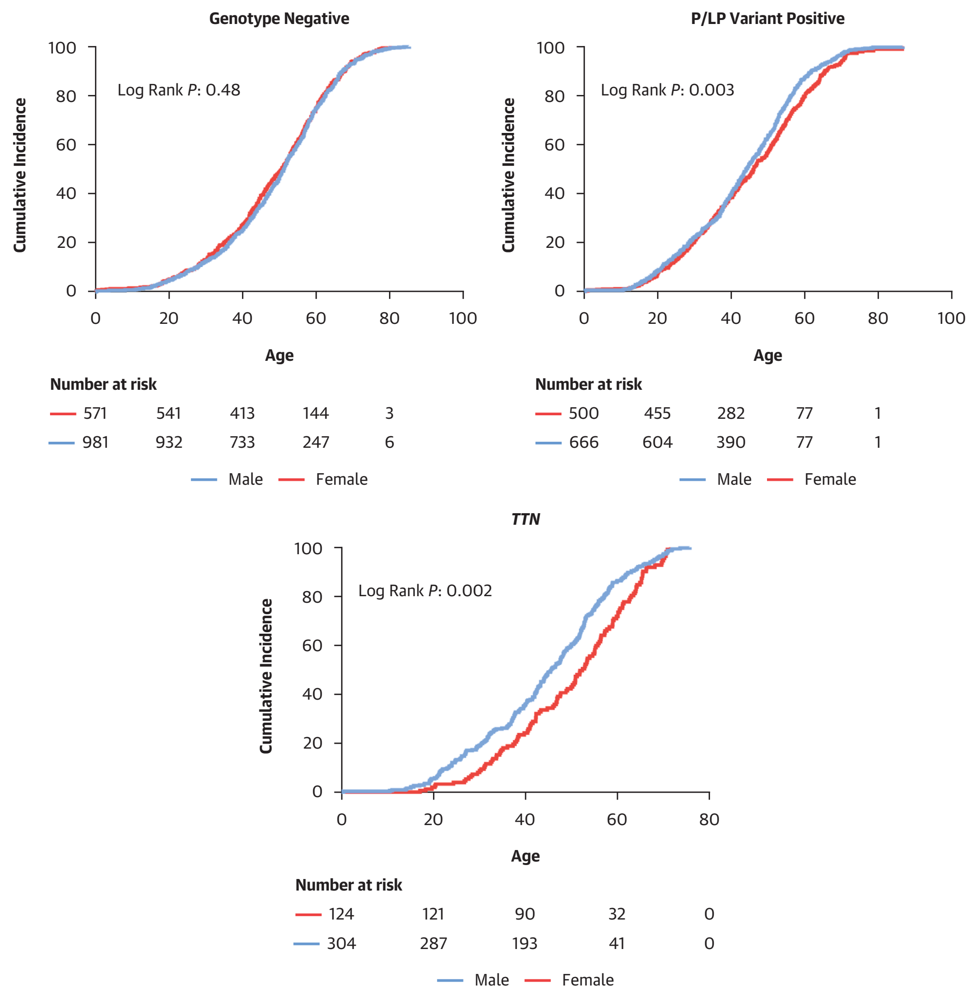
The figure represents data from the 6 most common DCM/ACM genes in the cohort with non-*TTN* sarcomeric genes depicted collectively. Males with *TTN* truncating variants (TTNtv) have earlier disease onset compared with females with TTNtv. There are no sex differences in age at disease onset for the other gene groups. The plots display the median and interquartile range, and each point on the plot represents data from an individual patient. Abbreviations as in **Figure 1**.

Overall, pediatric age at diagnosis demonstrated a bimodal distribution, with peaks in infancy and adolescence across both the P/LP− and P/LP+ groups (**Figure 7**), supporting the presence of distinct early- and late-presenting pediatric subgroups. The yield of P/LP variants increased with age, ranging from 22% to 32% in children diagnosed before 11 years of age (4 of 18 [22%] in late childhood, 9 of 28 [32%] in infancy) to 53% (60 of 114) in adolescents 11 to 18 years of age (**Figure 7**, Supplemental Table 5). **Figure 7** further highlights significant age-related differences in genotype composition: non-*TTN* sarcomeric variants were enriched in infancy and early childhood (7 of 9 [78%] and 3 of 4 [75%] of P/LP-positive cases, respectively), whereas this proportion declined in late childhood (1 of 4 [25%]) and adolescence (12 of 60 [20%]).

DISCUSSION

Using the SHaRe cohort of 3,410 adult and 174 pediatric genotyped DCM/ACM patients, we identified distinct sex-specific differences in genetic architecture at the gene level (**Central Illustration**). Gene-by-sex patterns for *TTN* and *DSP* confirm and extend prior population-based observations[9] in a large, clinically ascertained, multicenter registry and are not explained solely by differences in disease stage at presentation. Importantly, the present study adds several novel contributions, including inclusion of ACM, ARVC, and NDLVC phenotypes, broader gene coverage, life-course analyses using age-at-diagnosis and cumulative-incidence methods, and pediatric-onset genetic architecture. *TTN*-associated DCM/ACM was more common in males, whereas *DSP*-associated and sarcomeric variant-associated DCM/ACM were more frequent in females. Aside from *TTN* carriers with earlier disease onset in males, age at diagnosis in DCM/ACM was similar between sexes, suggesting that the male predominance is not driven by differences in disease stage across genotype-positive, genotype-negative, and VUS subgroups. Importantly, pediatric-onset DCM/ACM showed a bimodal age distribution, with peaks in infancy and adolescence and a distinct genetic architecture in early-onset disease characterized by enrichment of

FIGURE 4 **Cumulative Incidence of Diagnosis of DCM/ACM in Male and Female Patients**



Results are shown for the genotype-negative, P/LP variant-positive, and *TTN* groups. Kaplan-Meier curve relating age at diagnosis on the x-axis to cumulative incidence of DCM/ACM on the y-axis in male and female patients stratified by genetic status. Unaffected genotype-positive relatives were censored at last follow-up. Male patients have earlier disease onset in the P/LP group, which appears to be driven by earlier disease onset in males with TTNtv. Abbreviations as in Figures 1 and 3.

non-*TTN* sarcomeric variants, particularly among infants, revealing a novel and distinct genetic architecture in early-onset disease. These findings highlight gene- and sex-specific differences in the age at disease onset across both adult and pediatric populations, emphasizing the need for early genetic evaluation and tailored surveillance strategies, especially in pediatric patients.

SEX-BASED DIFFERENCES IN THE AGE AT DIAGNOSIS OF DCM AND ACM. The penetrance of variants in cardiomyopathy-associated genes is incomplete and age related, and expressivity is highly variable, features that present challenges for disease management.[13] Our analysis included both affected individuals and asymptomatic genotype-positive family members, using age at diagnosis as a practical surrogate for clinical expression, rather than aiming to estimate lifetime penetrance. Penetrance depends on gene and variant characteristics, as well as the modifying effects of genetic background and environment.[14] The impact of sex as a modifier of penetrance is incompletely understood.

These findings extend our previous report of sex differences in the genetic architecture of DCM by showing that the observed genetic patterns are not

TABLE 2 Baseline Characteristics of the Pediatric Cohort by Sex

	n	Male (n = 104)	Female (n = 70)	*P* Value
Age at diagnosis, y	174	12.5 ± 5.7	9.8 ± 6.8	0.006
ARVC diagnosis	17	10 (58.8)	7 (41.2)	0.77
Race	121			
White		64 (53)	43 (26)	
Other		0 (0)	3 (2.5)	
Black		5 (4.1)	3 (2.5)	
Asian		2 (1.6)	1 (0.8)	
Proband[a]	149	91 (87.5)	58 (82.8)	
Genotype	174			0.70
Genotype negative		37 (35.6)	21 (30.0)	
LP/P variant positive		45 (43.3)	32 (45.7)	
Multiple P/LP variants		1 (1.0)	0 (0.0)	
VUS		21 (20.2)	17 (24.3)	
TTE	150			
LVEDDi, mm/m^2		35.4 ± 18.6	44.7 ± 24.5	0.018
LVEDD z-score		1.8 ± 3.1	2.5 ± 4.2	0.33
LVEF, %		46.5 ± 14.5	45.4 ± 16.1	0.66
LVEF z-score		−4.4 ± 4.0	−4.4 ± 3.6	1.0
MRI	77			
LVEDVi, mL/m^2		119.8 ± 44.4	112.2 ± 64.0	0.60
RVEDVi, mL/m^2		94.5 ± 31.0	98.5 ± 58.8	0.75
LVEF, %		44.7 ± 11.7	50.8 ± 14.6	0.045
RVEF, %		47.3 ± 10.1	52.7 ± 10.2	0.030
Combined MRI or TTE LVEF, %	154	45.0 ± 13.9	46.4 ± 15.7	0.56

Values are mean ± SD or n (%), unless otherwise indicated. [a]Restricted to probands.
Abbreviations as in Table 1.

explained by differences in disease stage or clinical phenotype. Together, these studies indicate that biological sex modifies the age at disease onset of specific DCM/ACM genes rather than influencing disease risk solely through clinical or environmental factors.

In the present study, we find evidence for clear sex differences in the onset of disease of *TTN*, sarcomeric variants, and *DSP*. Conversely, sex does not appear to have a large modifying effect on the age at disease onset of other DCM/ACM genes in this cohort, though smaller effect sizes cannot be ruled out. The incorporation of age-at-onset and phenotype data represents a critical advance, allowing us to test whether sex differences reflect divergent disease trajectories or intrinsic molecular mechanisms. Our results suggest that sex influences gene-specific susceptibility, particularly within desmosomal pathways, independent of age at presentation or left ventricular function. This gene-by-sex interaction supports the hypothesis that sex-specific factors, potentially including hormonal, transcriptional, or immune modulation, alter myocardial response to genetic injury.

We do not imply that sex influences the inheritance of a disease-causing allele but rather that age at onset varies by sex differentially by disease gene. A number of factors may contribute to these sex biases in disease onset. They may reflect sex-based biases in diagnostic thresholds. In a hybrid study design using systematic review, meta-analysis, and primary analysis of UK Biobank data,[9] we recently showed that all-cause DCM, genetic DCM, and genotype-negative DCM were all twice as prevalent in males compared with females, and implementing sex-specific diagnostic thresholds on CMR reduced the male preponderance. However, although sex-agnostic diagnostic criteria contributed to underdiagnosis in female patients, overall, DCM was still more prevalent in males even after adjusting for sex-specific diagnosis. Our study replicates this overall disease predominance in males but carries forward these observations by demonstrating the granularity at a gene-specific level, with different directions of effect for *TTN* (more males) compared with *DSP* and non-*TTN* sarcomeric variants (more females), for example. Critically, in combination with the clinical phenotypic data such as age at onset and degree of ventricular impairment, we are also able to demonstrate that these differences are not due to different stages of disease presentation.

Environmental modifiers are likely to contribute to the divergent sex distribution of *TTN* and *DSP*. In TTNtv, second-hit effects, such as alcohol use or increased hemodynamic load, are well documented, differ by sex, and may contribute to higher disease expression in males.[15-17] Given that TTNtv-related cardiomyopathy often responds favorably to mechanical unloading,[18] the observed male predominance may in part reflect higher average BMI and associated load in males compared with females.

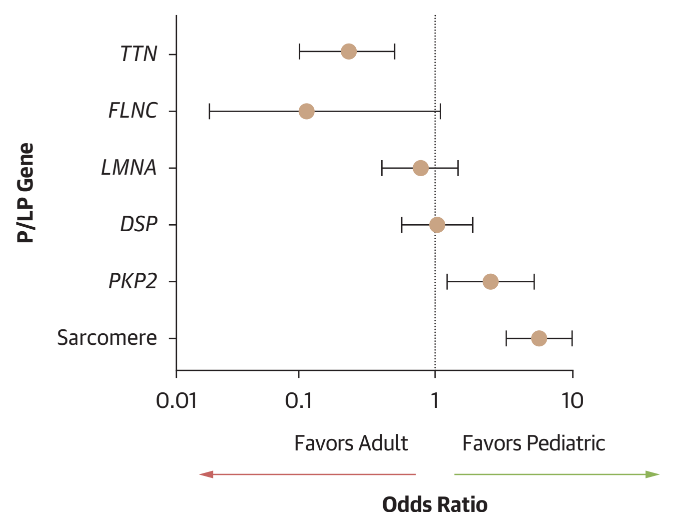
These mechanisms are less clear for *DSP*, for which sex-specific immune factors may drive the higher disease burden seen in females. Sex-related differences in immune function are well recognized, with systemic autoimmune diseases more prevalent in females. Recent research links this disparity to Xist expression, which reprograms T and B cells toward female-like immune profiles in males.[19] Notably, sex also influences the clinical expression of *DSP* cardiomyopathy. Female sex has been identified as a strong independent risk factor for ventricular arrhythmias, as shown in a validated, multivariable, *DSP*-specific risk score.[20] Sex biases in immunity may therefore contribute to the observed effects of female sex in *DSP* cardiomyopathy, but this remains speculative at present.

We observed that non-*TTN* variants in the genes that encode the sarcomeric thick and thin filaments were more common in females which is in contrast to sarcomeric HCM, which is generally more penetrant in males with earlier disease onset.[21] These opposing patterns suggest that although HCM and DCM share overlapping genetic substrates, the downstream phenotypic expression of these variants may be influenced by distinct, and potentially sex-modulated, mechanisms. This aligns with emerging evidence that shared genetic pathways in HCM and DCM may operate in opposite directions of effect, though the biological basis for these differences remains to be fully elucidated.[22]

Future research should prioritize investigating the role of sex hormones, reproductive factors, and environmental influences to better understand sex differences in gene-specific DCM/ACM. Sex hormones significantly affect myocardial remodeling, inflammation, and calcium handling, with estrogen generally exerting cardioprotective effects, while testosterone has been associated with increased hypertrophy, adverse remodeling, and a higher arrhythmic risk.[23] Improved understanding of these mechanisms may help explain gene-by-sex differences in disease expression and inform future studies evaluating whether surveillance strategies could ultimately be refined by age, sex, and genotype.

AGE-BASED DIFFERENCES IN PEDIATRIC DCM/ACM ONSET. We show that pediatric-onset DCM/ACM differs significantly from adult-onset cases, with a distinct genetic architecture and clinical presentations across different pediatric age groups. DCM/ACM diagnosis in pediatric patients follows a bimodal distribution, peaking in infancy and adolescence. By integrating pediatric and adult disease within the same multicenter registry framework, our findings provide a life-course perspective and reveal genetically distinct early and late onset pediatric subgroups. Our findings show a difference in the genetic causes of early and late pediatric DCM/ACM, paralleling the bimodal distribution of clinical presentation, and pediatric DCM/ACM was more common in males. Unlike adults, however, in whom *TTN* variants are the predominant genetic contributor, pediatric patients, particularly those presenting in infancy and early childhood, show a higher prevalence of sarcomeric gene variants. Ware et al[24] identified *MYH7* as the most common genetic cause of pediatric DCM in a cohort with predominant diagnosis in infancy (median age at diagnosis 1 year). In our study, with a broader age range of pediatric DCM/ACM diagnosis, the role of non-*TTN* sarcomeric variants was shown to be predominant in infant-onset and early childhood-onset DCM, whereas nonsarcomeric causes emerge in adolescent-onset disease. The over-representation of non-*TTN* sarcomeric variants in infant cases likely reflects the substantial impact these variants have on essential contractile proteins, leading to early and severe disruption of sarcomere function. Compared with TTNtv, which often require additional modifiers for disease expression, non-*TTN* sarcomeric variants may exert a stronger primary pathogenic effect, contributing to earlier disease onset. In contrast, the genetic architecture of DCM/ACM presenting in later childhood approximates that of adult DCM/ACM. This suggests that these cases represent severe versions of the adult phenotype with early disease onset. Further prospective studies will be needed to clarify the clinical trajectories and outcomes associated with these pediatric genetic subgroups and to better understand the mechanisms driving age-dependent disease expression.

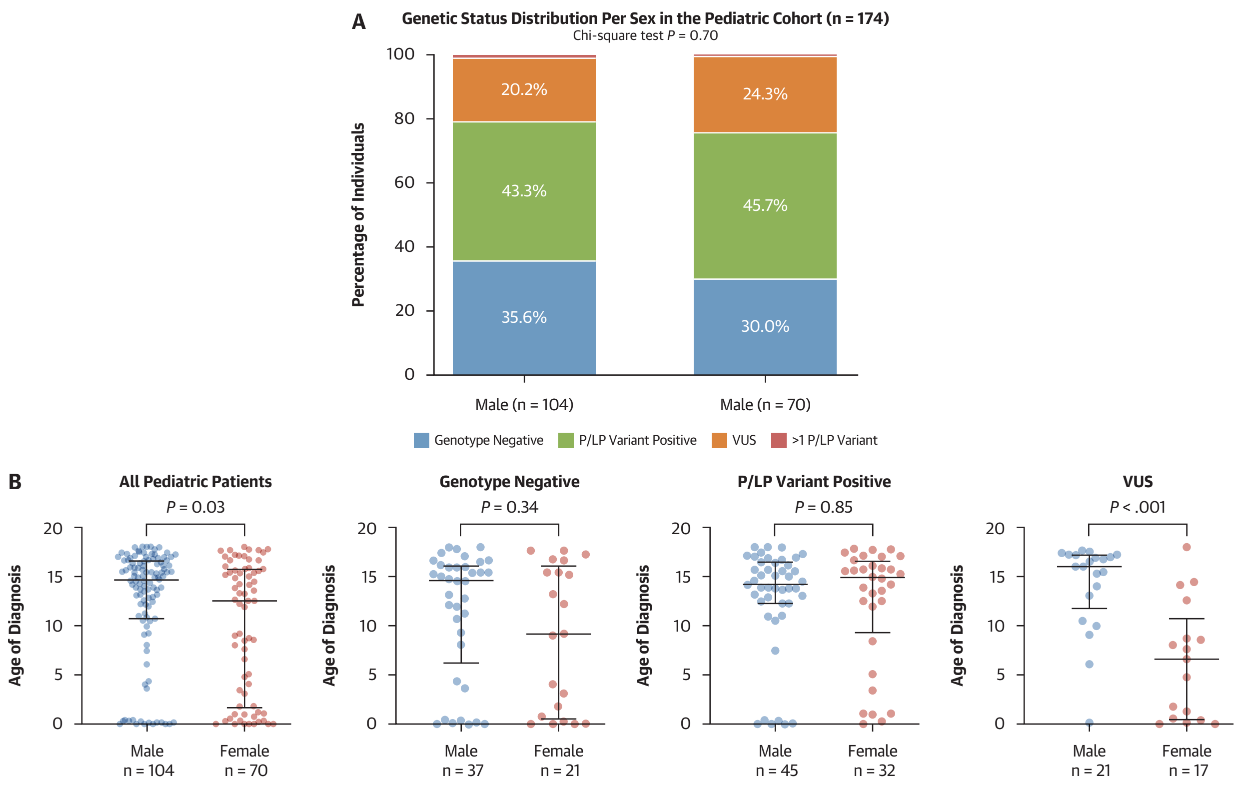
FIGURE 5 Gene-Specific Sex Bias Comparing the Adult and Pediatric Cohorts



ORs and 95% CIs for the distribution per genotype in adult and pediatric DCM/ACM patients. The dotted vertical line at an OR of 1 represents no age-at-onset bias. Genes positioned to the left of the line favor adults, while those to the right favor pediatric-onset disease. *TTN* cardiomyopathy is more likely to be found in adult-onset disease, whereas sarcomeric variants are more likely in pediatric-onset disease. Abbreviations as in Figure 1.

CLINICAL IMPLICATIONS. Our findings demonstrate that both sex and age influence the genetic architecture and clinical expression of DCM/ACM. The application of genetic testing in the care of patients

FIGURE 6 Sex and Age at Onset Across Genetic Subgroups in the Pediatric DCM/ACM Cohort

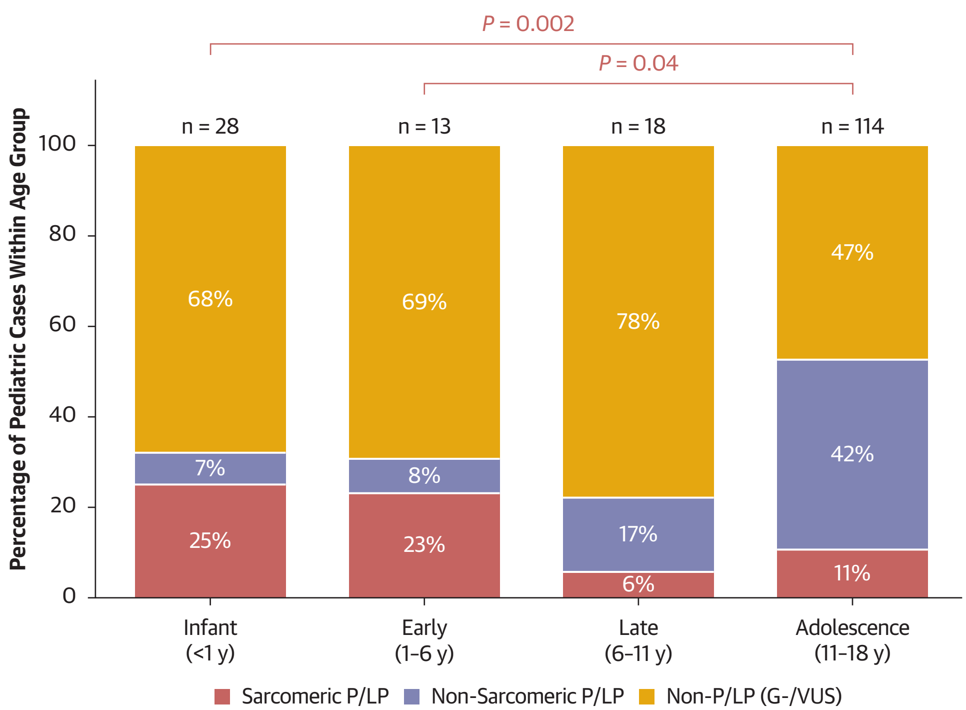


(A) Distribution of genetic status per sex in pediatric DCM/ACM patients in percentages. The chi-square test was used for statistical comparisons. (B) Age at DCM/ACM diagnosis in pediatric patients per genetic status. The plots display the median and interquartile range, and each point on the plot represents data from an individual patient. Abbreviations as in Figure 1.

with DCM/ACM has largely focused on improving cascade screening of at-risk relatives and refining arrhythmic risk stratification.[25] Longitudinal cardiac surveillance remains essential for genotype-positive individuals of both sexes. Our findings may refine counseling discussions by allowing clinicians to frame expectations regarding sex- and age-dependent disease expression in a gene-specific context. Knowledge of these gene-specific patterns may heighten clinical awareness when evaluating genotype-positive relatives. For example, the earlier disease expression observed in TTNtv carriers may inform counseling and clinical awareness when evaluating male carriers during longitudinal follow-up, whereas the relative enrichment of DSP and non-*TTN* sarcomeric variants in females underscores that female sex does not confer protection. Importantly, these findings should not be interpreted as supporting reduced surveillance in women, as a substantial proportion of affected individuals across genotypes are female. These findings should drive future work to evaluate drivers of these differences in disease expression and whether surveillance can be refined by age, sex, and genotype.

Current guidelines from the Heart Failure Society of America and the European Society of Cardiology recommend genetic testing and family screening for inherited cardiomyopathies but do not provide age-stratified screening recommendations according to disease gene-specific risks.[26,27] In this context, the enrichment of non-*TTN* sarcomeric variants among pediatric-onset cases in our cohort highlights the importance of heightened clinical vigilance in families with known sarcomeric cardiomyopathy, particularly in infancy and early childhood.[28] In

FIGURE 7 Distribution of Pediatric Dilated Cardiomyopathy/Arrhythmogenic Cardiomyopathy Diagnoses by Age Group and Genotype Class

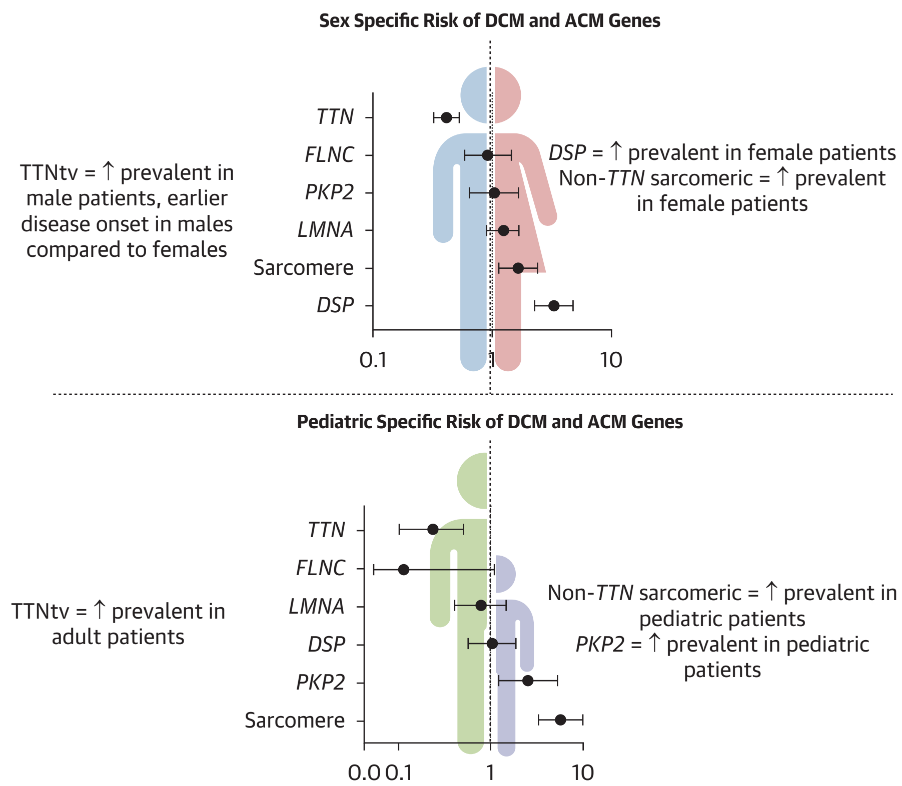


Bars represent the percentages of cases per age bin, with segments indicating sarcomeric P/LP, nonsarcomeric P/LP, and non-P/LP (genotype-negative [G−] or VUS). Total n is displayed above each bar. Brackets indicate Fisher exact test comparisons of the proportion of sarcomeric P/LP variants among P/LP-positive cases in each age group vs adolescence (reference); only statistically significant comparisons are shown. Abbreviations as in Figure 1.

selected circumstances, fetal echocardiography has been reported in families with severe sarcomeric cardiomyopathy, as emerging evidence suggests that disease may manifest prenatally.[29-31] Although the logistical challenges of cardiac imaging in early childhood are considerable, the consequences of a missed diagnosis of DCM in infancy are substantial.[32] Integration of these gene- and sex-dependent patterns into clinical practice may enable earlier diagnosis, more precise risk stratification, and optimized management across the lifespan.

STUDY LIMITATIONS. Centralized adjudication of genetic variants was not performed, but all classification followed American College of Medical Genetics and Genomics frameworks. In this registry-based cohort, age at diagnosis represents a pragmatic, clinically observable measure of disease expression. Because this is a clinically ascertained registry of affected individuals, gene and variant frequencies reported here should not be interpreted as reflecting population prevalence. Cumulative-incidence analyses incorporating relatives may be affected by variation in cascade screening and follow-up intensity (including sex- or age-related differences) and therefore reflect clinical diagnosis under surveillance rather than formal lifetime penetrance. The genetic architecture of pediatric DCM/ACM may be influenced by ascertainment bias, as some patients were recruited from adult centers despite childhood onset, and cases leading to early death or transplantation are likely under-represented. Information on family history and de novo status was not systematically available across SHaRe sites and could not be assessed in the pediatric-onset subgroup. The referral-based nature of SHaRe may also limit generalizability to the broader disease population. Nevertheless, this remains one of the largest genotyped and phenotyped DCM/ACM registry

CENTRAL ILLUSTRATION Sex- and Age-Specific Genetic Risk for Dilated and Arrhythmogenic Cardiomyopathy: Insights From SHaRe



Stroeks SLVM, et al. JACC. 2026;■(■):■-■.

SHaRe = Sarcomeric Human Cardiomyopathy Registry; TTNtv = *TTN* truncating variant.

available. The pediatric cohort (n = 174) represents a meaningful sample size in this rare disease context, comparable with recently published studies.[33]

Small subgroup sizes restrict the precision of gene-specific estimates, and we could not account for unmeasured confounders such as BMI, hypertension, diabetes, and other environmental factors. Race/ethnicity was available only for a subset of participants; among those with available data, most were White, which limits ancestral diversity and may constrain the generalizability of these findings. Finally, the exclusion of 1,406 individuals without age at diagnosis may have introduced selection bias, although the remaining cohort was large and well characterized.

Future studies should investigate modifiers of disease onset and expression to better understand the gene-by-sex differences observed in this study. In particular, the biological mechanisms underlying these patterns remain incompletely understood, including the female predominance observed in DSP-associated cardiomyopathy. Potential contributors may include differences in sex hormone signaling, immune regulation, myocardial remodeling pathways, and genetic background, as well as environmental exposures that may interact with genotype to influence disease onset and severity. Investigation of hormonal modifiers across the life course, including pregnancy, pregnancy complications, lactation, and exposure to hormonal contraception, may provide further insight into how hormonal fluctuations influence DCM/ACM risk and disease expression in genetically susceptible individuals. Population-based cohorts and large-scale biobank resources with linked genomic and longitudinal health data will be particularly valuable for estimating age- and sex-dependent penetrance of cardiomyopathy-associated variants in unselected populations, helping disentangle true biological differences from ascertainment bias inherent to clinically referred registries. Integration of genetic data with detailed phenotyping, reproductive history, environmental exposures, and longitudinal imaging may provide further insight into modifiers of disease onset and progression. Finally, prospective multicenter cohorts with long-term follow-up of genotype-positive relatives will be essential to better define gene-, age-, and sex-specific disease trajectories and to determine whether the observed gene-by-sex differences ultimately justify refinement of surveillance intervals or screening strategies in clinical practice.

CONCLUSIONS

In this registry study of genotyped DCM/ACM patients including 3,410 adult and 174 pediatric patients, there was a greater male predominance for *TTN* with earlier disease onset in *TTN* cardiomyopathy in males and greater female predominance for *DSP* and non-*TTN* sarcomere variants. Pediatric-onset DCM/ACM is caused predominantly by non-*TTN* sarcomeric variants, especially during infancy. These findings may help inform future guideline development for disease surveillance in adults and children across both sexes with DCM/ACM and prioritize future research into the mechanisms of observed sex-based differences.

What This Study Demonstrates

- In a large multicenter registry of genotyped DCM and ACM, gene-specific sex differences were observed across the disease spectrum.
- TTNtv were more common and associated with earlier disease expression in males compared with females, whereas DSP and non-*TTN* sarcomeric variants were relatively enriched in females compared with males.
- Aside from *TTN*-associated disease, age at diagnosis was largely similar between sexes, suggesting that the overall male predominance in DCM/ACM is not explained solely by differences in disease stage at presentation.
- Pediatric-onset DCM/ACM demonstrated a distinct genetic architecture with enrichment of non-*TTN* sarcomeric variants in infancy and a bimodal age distribution with peaks in infancy and adolescence.

What This May Mean for Clinicians

- Longitudinal cardiac surveillance remains essential for genotype-positive individuals of both sexes, as a substantial proportion of affected individuals are female across genotypes.
- Gene- and sex-specific patterns of disease expression may help refine counseling of genotype-positive relatives and frame expectations regarding potential age at disease onset.
- Clinicians should remain vigilant in women with DSP and non-*TTN* sarcomeric variants, recognizing that female sex does not confer protection from disease expression.
- In families with pathogenic non-*TTN* sarcomeric variants, heightened clinical awareness in early life may be warranted given the enrichment of pediatric disease in infancy and early childhood.

What Remains Uncertain Pending Prospective Validation

- The biological mechanisms underlying gene-by-sex differences in DCM/ACM, including the observed female predominance in DSP cardiomyopathy.
- How hormonal (including life-course hormonal exposures such as pregnancy and hormonal contraception), environmental, and genetic modifiers interact to influence disease onset and severity.
- Whether the observed gene- and sex-specific patterns ultimately justify refinement of surveillance intervals or screening strategies in clinical practice.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Ware was supported by the Medical Research Council (UK), British Heart Foundation (RE/24/130023), Sir Jules Thorn Charitable Trust (21JTA), and the NIHR Imperial Biomedical Research Centre. Dr Tayal has received funding from the Medical Research Council (MR/W023830/1), the Royal Society (RG\R2\232551), and the National Institute for Health Research Imperial Biomedical Research Centre; and has received infrastructure support from the British Heart Foundation (RE/24/130023). Dr Lakdawala has received unrestricted research support from the O'Hare Family Foundation and the Steggall Family Foundation. Drs Taylor and Mestroni have received grant support from the National Institutes of Health (grants X01 HL139403, UL1 RR025870, R01HL164634, R01HL147064, and R01HL170012). Dr Halliday is supported by the British Heart Foundation (grant FS/ICRF/24/26128) and the Rosetrees Trust. Dr Bart was supported by a Fullbright Future scholarship. The funders did not have a role in the study design; in the collection, analysis, or interpretation of data; in the writing of the report; or in the decision to submit the article for publication. Dr Wheeler has received research support from AstraZeneca, BMS, Cytokinetics, Salubris Bio, and Nihon Kohden; and has received modest consulting fees from Cytokinetics, and Braveheart Bio. Dr Ware has received research support from Bristol Myers Squibb; acts as a paid advisor to Health Lumen, Tenaya Therapeutics, and Solid Biosciences; and is a founder with equity in Saturnus Bio. Dr Tayal is a research editor at the *BMJ*; has consulted for Kardigan; and is supported by a grant from the Asmarley Charitable Trust. Dr Lakdawala has received unrestricted research funding from Bristol Myers Squibb and Pfizer; and has received modest consulting incomes from Alexion, Bristol Myers Squibb, Pfizer, Cytokinetics, Tenaya, Nuevocor, Bayer, and Lexeo. Dr Bart has received research funding and speaking fees from Pfizer; and has received advisory board fees from Pfizer, BridgeBio, Novo Nordisk, and Bristol Myers Squibb. Dr Mestroni has received advisory board fees from Tenaya Therapeutics and Rocket Biopharma; and has received research funding from Pfizer, Alexion, and Greenstone Bio. Dr Halliday has consulted for AstraZeneca and Zoll. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

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KEY WORDS dilated cardiomyopathy, genetics, pediatrics, penetrance, sex

APPENDIX For supplemental tables and supplemental figures, please see the online version of this paper.